Midterm 1 Practice Problems 1

This is very roughly the level and quantity of problems you can expect on the midterm.

Practice 1.1

For each statement in the list below, explain whether the statement is true or false, and prove your answer. For false statements make sure to provide a counterexample, not just an explanation why a proof one might try won't work.

Wherever the symbol X appears, it denotes a metric space. Unless otherwise stated, assume the usual distance function on $\mathbb{R}$.

(1) Every subset of $\mathbb{Q}$ which is bounded below in $\mathbb{Q}$ has a greatest lower bound in $\mathbb{R}$.

(2) If $E \subset \mathbb{R}$ is bounded, then

$$\{a + b \mid a, b \in E\}$$

is also bounded.

(3) The ordered set $\mathbb{Z}$ has the least upper bound property.

(4) The set of all functions $f : \{1, 2, 3\} \to \mathbb{N}$ is countable.

(5) If $E \subset \mathbb{R}$ is countable, then E is closed.

(6) If $E \subset X$, and $\bar{E} = X$ then E is dense in X.

(7) Given any collection of closed intervals in $\mathbb{R}$, the union of the collection is closed.

(8) If every point of $E \subset X$ is a limit point of E, then E is closed.

Practice 1.2

Consider the set

$$S = \{0, 1, \ldots, 14\}.$$

We can equip S with addition and multiplication rules by using the usual rules for integers modulo 15 (so, for example, $6 + 11 = 2$, $4 \cdot 2 = 8$, and $6 \cdot 7 = 12$).

Prove that, with these addition and multiplication rules, S cannot be made into a field.

Practice 1.3

Suppose that S is an ordered set, and $A \subset S$ has a maximum element y. Prove that y is also a supremum of A.

Practice 1.4

Show that $\mathbb{R}$ is not compact by giving an open cover and showing it has no finite subcover.

Practice 1.5

Let X be a metric space, $p \in X$ be a point, and $r > 0$. Show that the closed ball $\{q \in X | d(p, q) \leq r\}$ is closed.